TileSmith Mobile

LED Wall Pixel Map & Test Pattern Generator · Android and iOS app manual

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Lay out LED walls, test patterns and pixel maps, plan data and power wiring, and watch an animated technician build the wall. On your tablet or phone, offline, with files and videos you can share straight from the device.

This manual covers the mobile app. The Windows and Mac desktop app has its own manual (TileSmith-Manual.pdf).
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1. The mobile app

TileSmith Mobile is the TileSmith LED wall tool on Android and iOS tablets and phones. It runs the **same TileSmith page** as the desktop and web versions, so the canvas, walls, tile models, test patterns, engineering, wiring, power and the build animation are identical: the same layout gives the same picture and the same numbers on a tablet as on the desktop. Everything works offline and is saved on the device automatically.

	Mobile app	Desktop app (Windows, Mac)
Canvas, walls, presets, patterns, engineering, wiring, power, build animation, DVD screensaver	Yes	Yes
Export PNG, PDF, Resolume XML	Yes: through the share sheet (Files, Drive, AirDrop, e-mail ...)	Yes: saved to folders
Export video	H.264 MP4, recorded in real time with the device's encoder	HAP, H.264, H.265, ProRes 422 HQ, ProRes 4444 (ffmpeg)
Video export: DVD logo (perfect loop video)	No	Yes
NDI® output, Spout (Windows), Syphon (Mac)	No (buttons and menus hidden)	Yes
Touch: pinch to zoom, two-finger pan	Yes	Windows touch screens

> Tablets give the most room. On a phone the panels stack under each other and you scroll between them (chapter 3). On a phone the viewport sits at the bottom of the page (chapter 3).

2. Installing

TileSmith Mobile is built from the TileSmith project. It is not in the app stores yet.

Android (Android 7.0 or newer)

- **From the project's mobile folder** (Capacitor): follow **BUILD-ANDROID.txt**. You need Node.js and Android Studio; plug in your device and press Run.
- **From Google AI Studio**: follow **00-START-HERE.md** in the AI Studio kit. AI Studio builds a small Android app around the same TileSmith page, and it can install it on a USB-connected phone from the browser. The kit's `android-project` folder is the same app, ready for Android Studio.

Both builds behave as this manual describes. They have different app IDs, so you can install both side by side and compare them. For video recording, keep **Android System WebView** and **Chrome** up to date from Google Play.

iPhone and iPad (iOS / iPadOS 15 or newer)

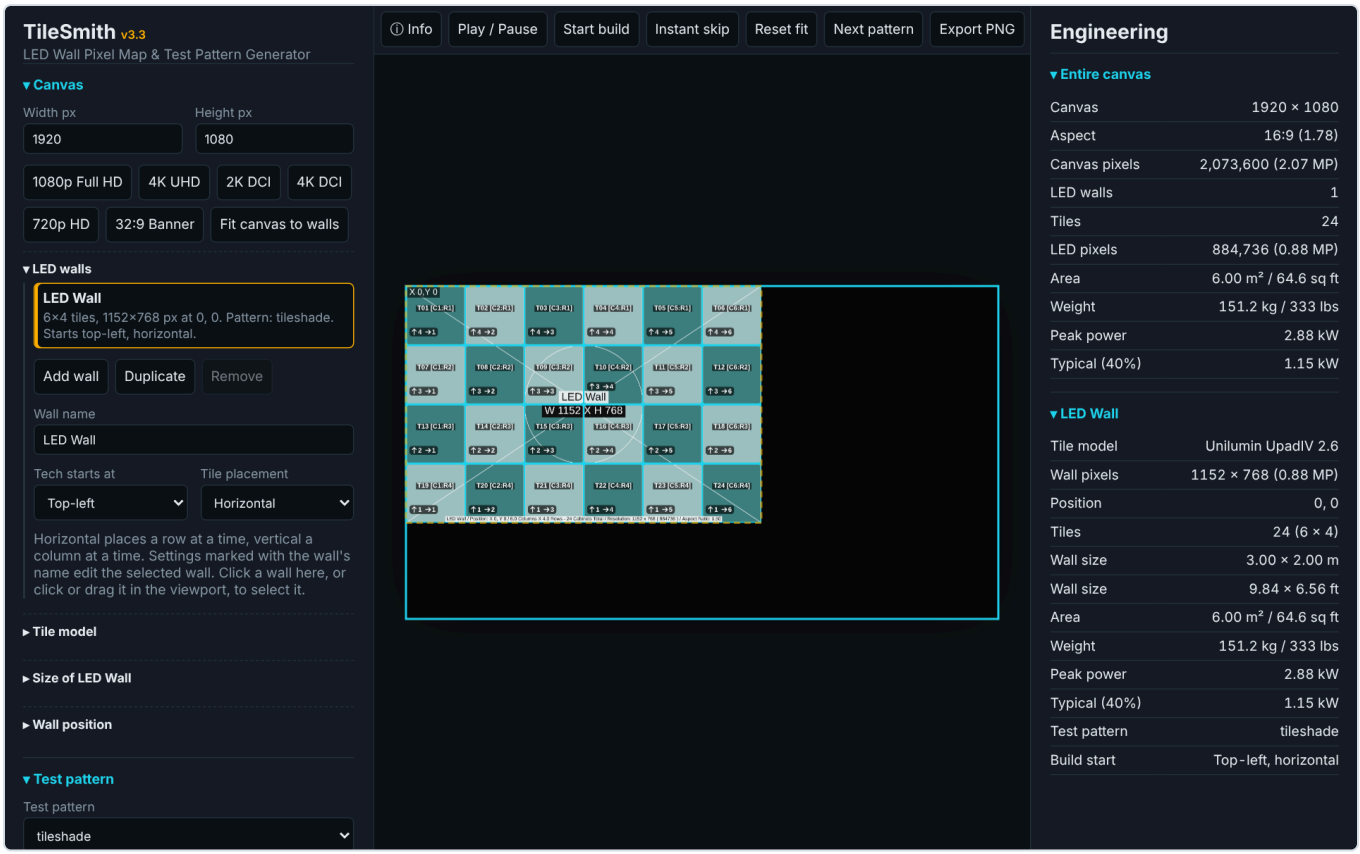
Follow **BUILD-IOS.txt**. You build on a Mac with Xcode and sign with your Apple ID. With a free Apple ID the app runs for 7 days, then needs installing again from Xcode; the paid Apple Developer Program removes that limit and adds TestFlight.

The header shows the version (**v3.5**), so you can check which build you are running.

3. The screen

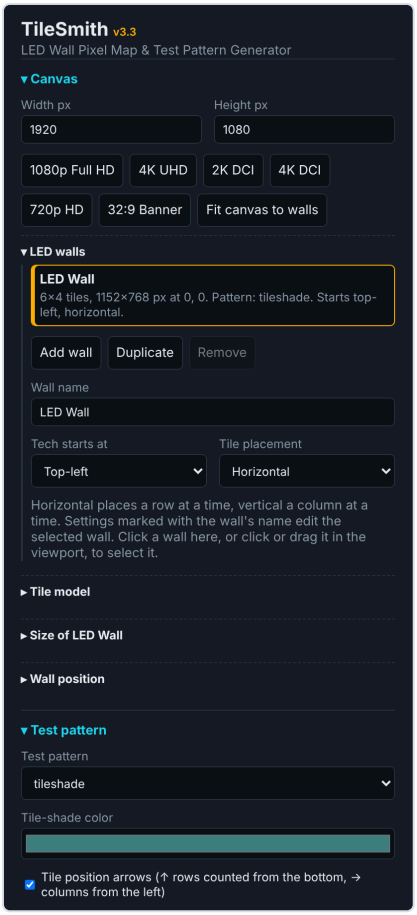
On a **tablet**, or any screen wider than 900 points, TileSmith shows three columns:

- **Settings** on the left, in collapsible sections. It scrolls on its own.
- The **top bar** of actions and the **viewport** in the middle. The viewport is a preview of the master canvas: the picture your LED processor or media server would get.
- **Engineering data** on the right.

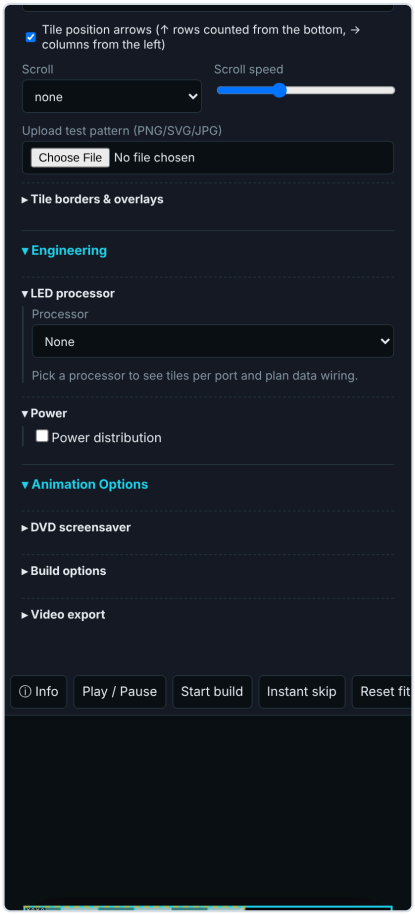


Tablet, landscape: a new install shows one 6 × 4 wall on a 1920 × 1080 canvas.

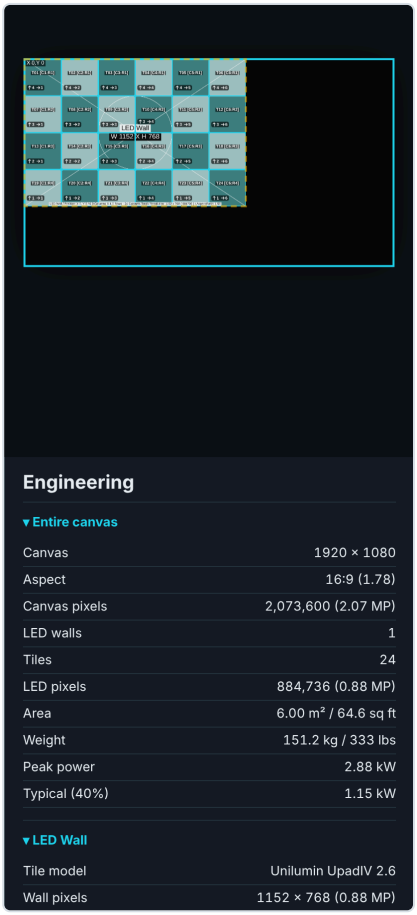
On a **phone** the page runs top to bottom: the settings, the collapsed **Engineering data** menu, the button bar (wrapped onto several rows, ending with **Start LED Build** and **Start DVD Screensaver**) and the viewport at the bottom, with a **Zoom** slider along its lower edge. With several canvases a phone shows them one above the other. Turning the phone sideways shows the three columns, side by side but narrow.



Phone, screen 1

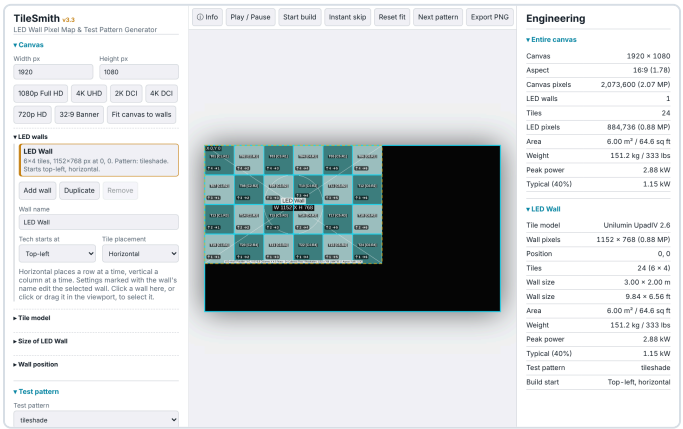


Phone, screen 2

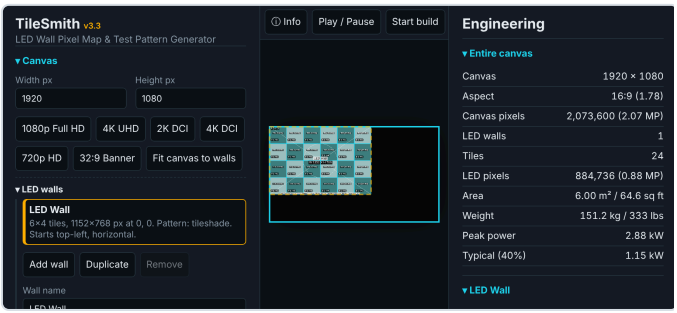


Phone, screen 3

The app follows the system **light or dark** setting.

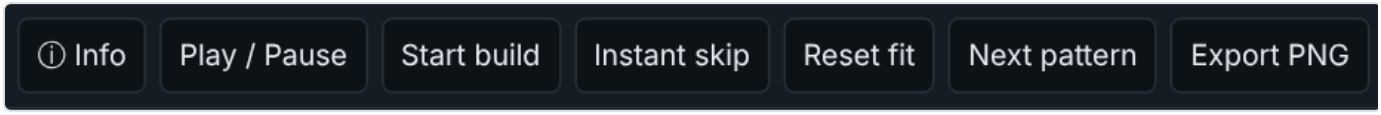


Light theme



Phone in landscape

Top bar



Top bar. It scrolls sideways when it does not fit.

Button	What it does
ⓘ Info	About TileSmith, credits, contact, Ko-fi and Reset saved settings (chapter 16).
Save and **Open**	Next to New Project. Save keeps the whole project, with your images and crew, in one .tilesmith file; Open loads one.
Center canvas	Centres the canvas in the viewport.
Show entire canvas	Fits the whole canvas in the viewport again.
New Project	Asks first, then wipes everything and starts a blank canvas. Undo brings it back.
Undo	Opens a list of the last 20 changes; tap one to go back to just before it. The first item can be *Redo*.
Next pattern	Moves the selected wall to the next test pattern; a message at the bottom names it.
Export PNG / Export PDF	Save the canvas picture, or one PDF with the full raster, the spec sheet and the cable labels (chapter 14).
Resolume XML ▾	Saves a Resolume Arena Advanced Output preset: One slice per LED wall or One slice per tile.
Report a bug / feedback	At the end of the bar, next to ⓘ Info. Captures a screenshot, lets you mark it up and add a description, then saves the marked-up screenshot through the share sheet and opens an email to tilesmithapp@gmail.com with your description — attach the saved screenshot before sending.

Next to the buttons: a red recording status, the zoom level and how many frames per second the canvas draws (*still* when nothing moves).

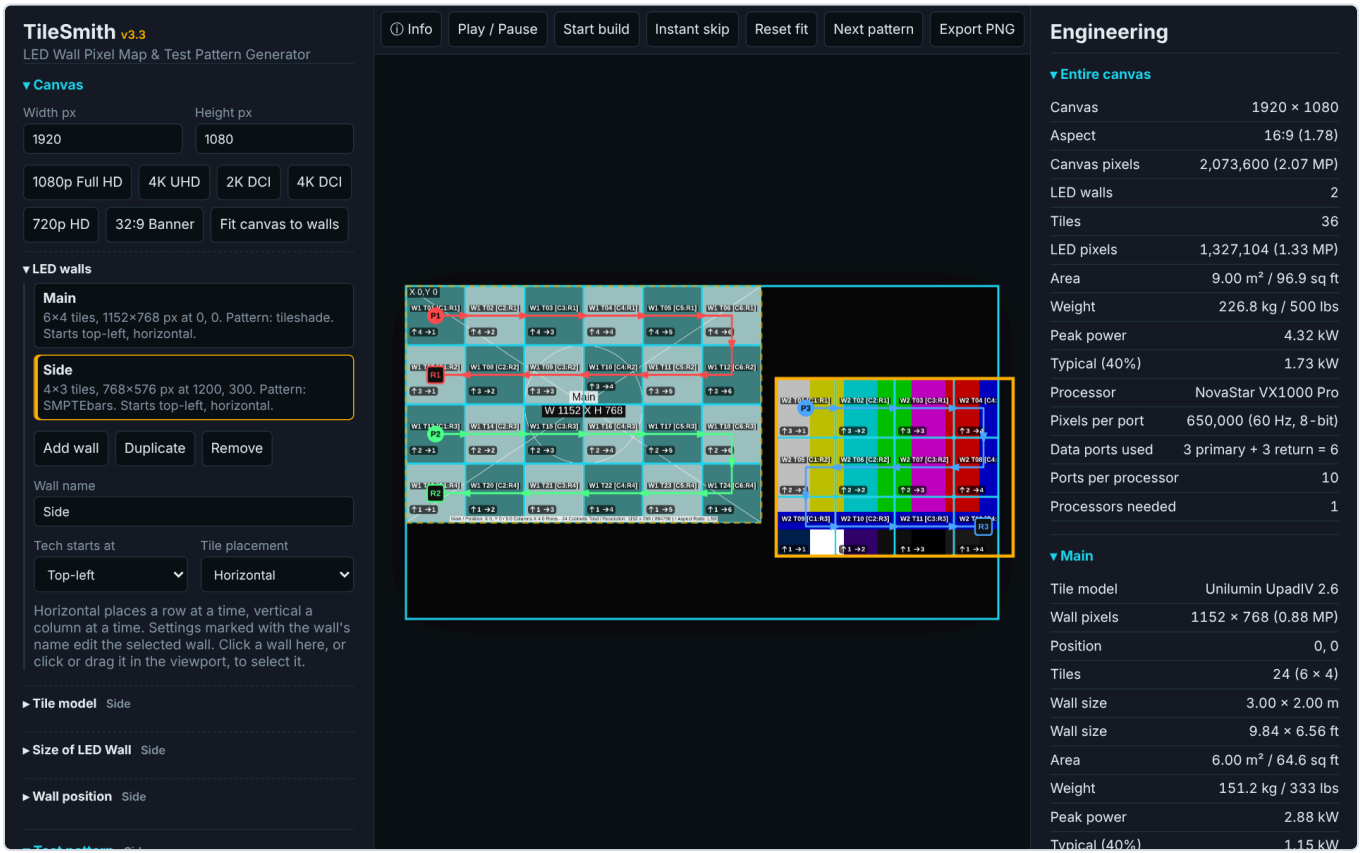
4. Touch controls

Tap	Buttons, checkboxes and lists, as usual. Tap a wall in the viewport to select it.
One finger on a wall	Drag it. It snaps to the canvas edges and to other walls, keeping a 2 px gap.
One finger, editing modes	In *Wire tiles*, *Remove tiles* and *Mark half tiles*, tapping and dragging edit the selected wall instead of moving it.
One finger on empty space	Pans the viewport.
Two fingers	Pinch to zoom (up to 500%) around your fingers, and move to pan. Only the viewport zooms, never the whole screen.
Scroll	Swipe the settings or Engineering data panel to scroll it. On a phone, swipe outside the viewport to scroll the page.
Show entire canvas	Brings the whole canvas back into view.

Text and number fields open the keyboard. Colour fields open the system colour picker. Upload fields open the photo picker or file browser. Web links, Ko-fi and e-mail open in your browser or mail app.

5. Quick start

1. Canvas: tap the canvas size your media server outputs, for example **1080p Full HD**.
2. Tile model and Size of LED Wall: choose your tile preset, then columns and rows.
3. Test pattern: choose a pattern, or tap **Next pattern** to step through them.
4. Engineering › LED processor: choose your processor, then a pattern under Data wiring.
5. Engineering › Power: tick **Power distribution**, then a pattern under Power wiring.
6. Open **Animation Options › LED build** and tap **Start** to watch the technician build the wall, or open **Video export** and tap **Export video** to record it.
7. Tap **Export PNG** or **Export PDF** and pick where the file goes.



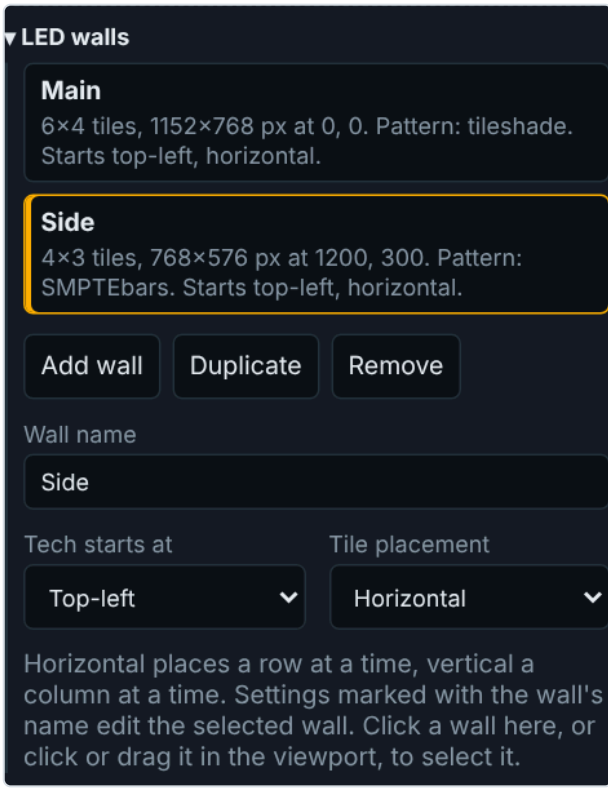
Two walls, a NovaStar VX1000 Pro and Snake data wiring.

6. Canvas and LED walls

Canvases lists every output raster; **Add canvas** makes another, each with its own submenu of LED wall options and its own outputs, shown side by side in the viewport (tap one to select it). Each canvas sets its output raster: Width and Height in pixels, or a preset (1080p Full HD, 4K UHD, 2K DCI, 4K DCI, 720p HD, 32:9 Banner). **Fit canvas to walls** moves the walls to the top-left and makes the canvas exactly as big as they are.

LED walls lists every wall with its size, pixels, position and pattern. Tap a wall to select it; the selected wall has an amber border. **Add wall** places a new 4 × 3 wall beside the others. **Duplicate** copies the selected wall, and **Remove** deletes it (one wall always stays).

Sections marked with a wall's name (for example *Tile model · Main*) change the selected wall only. **Tech starts at** and **Tile placement** set where the technician begins and whether he builds row by row (Horizontal) or column by column (Vertical).



LED walls



Three walls on one canvas, each with its own pattern. With several walls, tile IDs start with W1, W2 ...

Wall position

Type **Offset X** and **Offset Y** in canvas pixels, or tap an alignment button: Top-Left, Center Both, Top-Center, Bottom-Left, Bottom-Center. You can also drag the wall in the viewport.



Wall position

7. Tile model, size and shape

Tile model has a **Brand** menu (ROE, Absen, Unilumin, INFiLED, YES TECH, Owl Vision) and a **Model** menu for that brand. The model fills in the tile's pixels, millimetres, pitch, weight, max watts and depth. Change any of them and the model becomes **Custom**. Pick **Custom** in Brand, fill in the fields, name it and press **Save as my preset** to keep it for later; it is also saved inside any project that uses it.

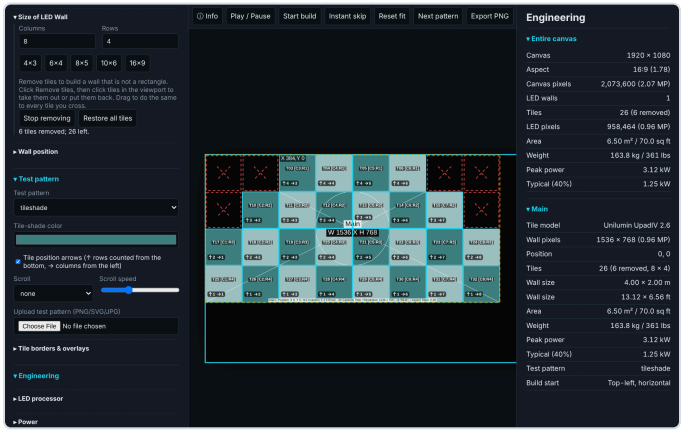
Size of LED Wall sets Columns and Rows, with quick buttons 4×3, 6×4, 8×5, 10×6 and 16×9.



Tile model

Removing tiles

For a wall that is not a rectangle, tap **Remove tiles** (Size of LED Wall). Then tap tiles in the viewport to take them out or put them back, or drag across several. Removed spots show a red ✕ in the viewport only, never in the output. **Restore all tiles** puts them all back.



Remove tiles mode



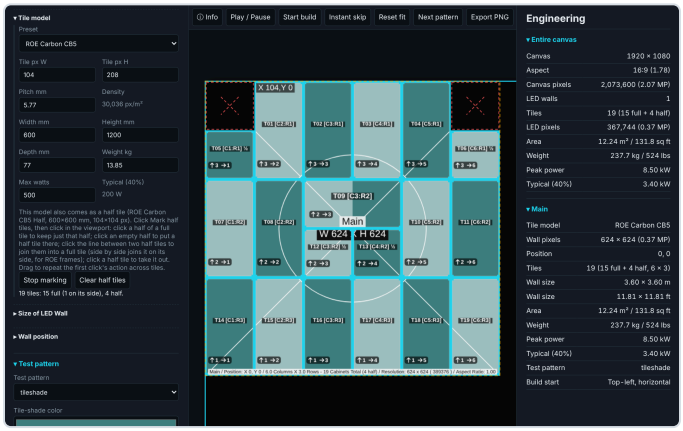
The output: six corner tiles removed

Half tiles

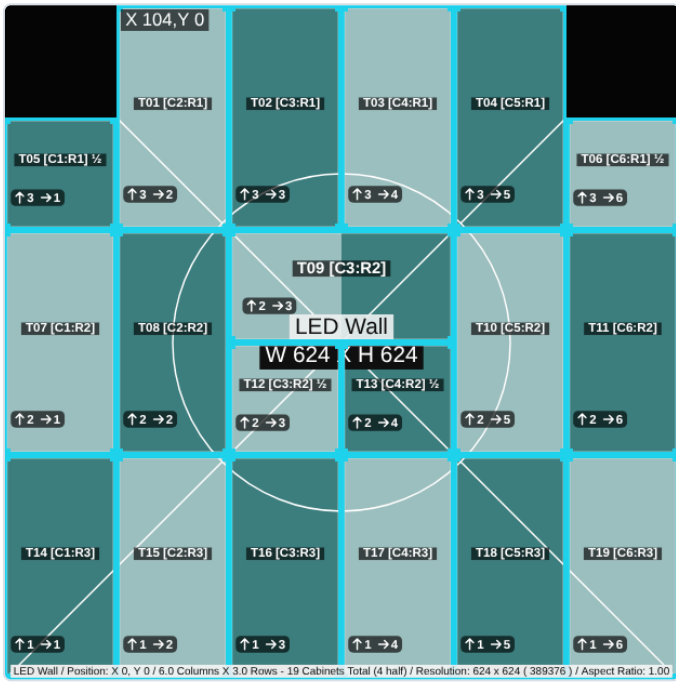
Some models also come as a half tile (ROE Carbon CB3/CB5/CB8, YES TECH MU 500×1000, YES TECH MT II 1000×500). For these, the **Size of LED Wall** submenu shows **Convert bottom row to half tiles** and **Convert top row to half tiles** (tap again to restore full tiles) and **Mark half tiles**. While marking, a red **STOP MARKING HALF TILES** button sits at the top of the viewport. Tap in the viewport:

- Tap one half of a full tile to keep just that half.
- Tap an empty half to put a half tile there.
- Tap the line between two half tiles to join them into a full tile. This works upright, offset by half a tile, or on its side (ROE frames).
- Tap a half tile to take it out.

Drag to repeat the first tap's action. **Clear half tiles** goes back to full tiles.



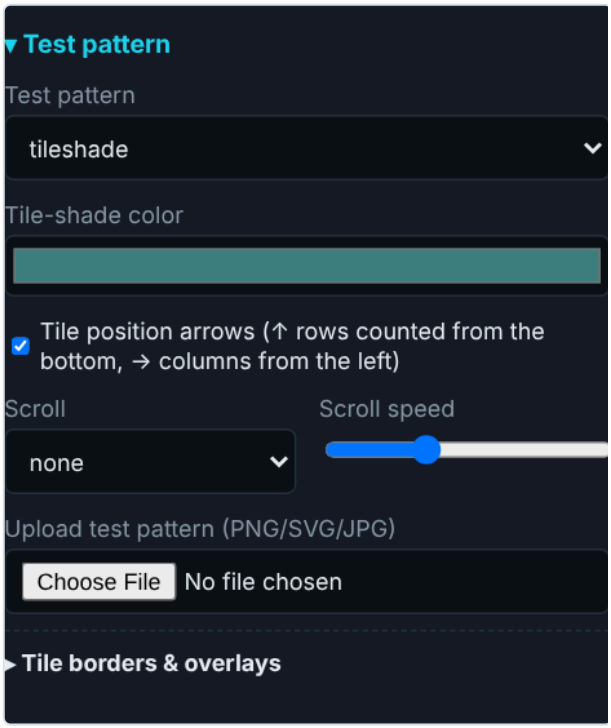
Mark half tiles (ROE Carbon CB5)



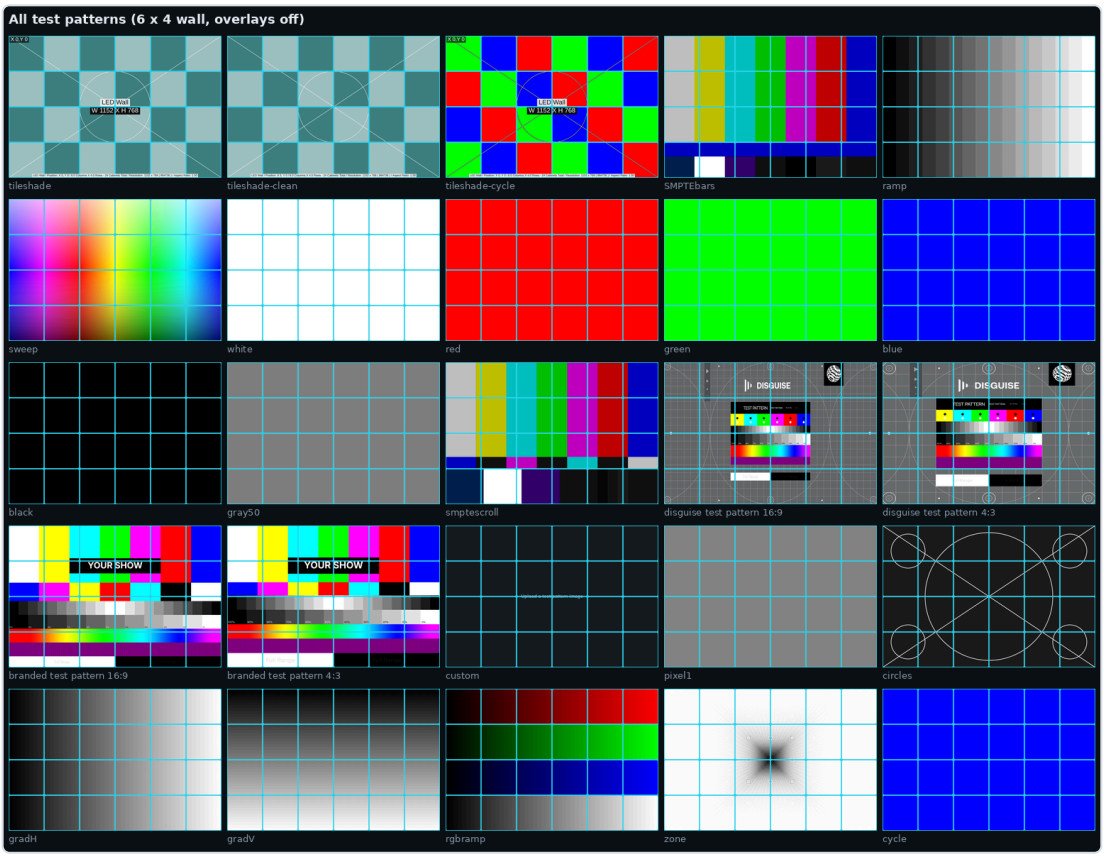
The output; half tiles end in ½

8. Test patterns and overlays

Choose the pattern for the selected wall under **Test pattern**. **Tile-shade color** appears for tileshade and tileshade-clean. For image patterns, **Image fit** decides how the picture fills the wall. **Scroll** moves any pattern diagonally, sideways or down, at **Scroll speed**. **Upload test pattern** opens the photo picker for the *custom* pattern. The branded patterns have a centre box for your own text or logo.



Test pattern



All 25 patterns (overlays off)

Pattern	What it shows
tileshade	Pixel map. Every tile is a light or dark shade of one colour (set it with **Tile-shade color**), with corner-to-corner diagonals, a centre circle, the wall name and resolution in the middle, the canvas position of the first tile's top-left corner in a see-through dark box, and a footer with the wall name, position, columns, rows, cabinet count, resolution and aspect ratio. This is the default pattern.
tileshade-clean	The same pixel map without the name, size and position boxes on the wall; that information is only in the footer.
tileshade-cycle	The tileshade layout with every tile stepping through red, green and blue. Each colour holds for a second, then the next one slides in from the left over 1.5 seconds. Each tile starts at a different point in the sequence, so a tile never shows the same colour as the tiles above, below, left or right of it. Good for spotting dead pixels, colour mismatches and swapped tiles. Tile-shade color is not used.
SMPTEbars	SMPTE colour bars image. Image fit and Scroll apply.
ramp	16-step greyscale from black to white, left to right.
sweep	Moving rainbow with a light-to-dark vertical fade (animated).
white	Full-field white.
red	Full-field red.
green	Full-field green.
blue	Full-field blue.
black	Full-field black.
gray50	Full-field 50% grey.
disguise test pattern 16:9	Built-in 1920 × 1080 image with grid, circles, greyscale steps and colour references.
disguise test pattern 4:3	Built-in 1024 × 768 version of the same image.
branded test pattern 16:9	Built-in 1920 × 1080 colour bars with a black centre box. Put your own text or logo in the box with **Center box**.
branded test pattern 4:3	Built-in 1024 × 768 version with the same centre box.
custom	Your own image: click **Upload test pattern**. Images keep their full size up to 8192 px. Image fit and Scroll apply.
pixel1	Single-pixel checkerboard (every other pixel on). Good for checking pixel mapping and receiving-card setup.
circles	Circles and crosshairs for checking geometry and aspect ratio.
gradH	Smooth black-to-white gradient, left to right.

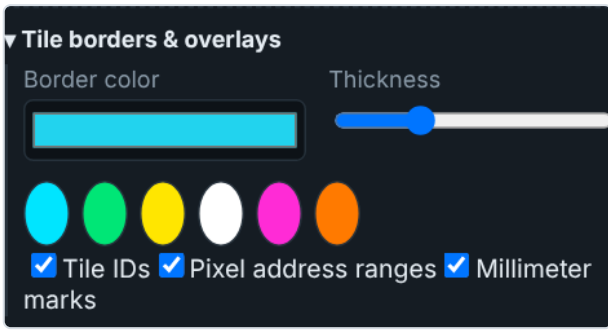
Pattern	What it shows
gradV	Smooth black-to-white gradient, top to bottom.
rgbramp	Four ramps from black to red, green, blue and white.
zone	Zone plate (concentric rings) for spotting scaling and moiré.
cycle	Full-field colour that changes every 1.5 seconds (animated).

Image fit	What it does
Stretch to wall	Fills the wall exactly. The image is stretched if its shape differs from the wall.
Fit inside wall	Shows the whole image without stretching; black bars fill the gaps.
Fill wall (crops edges)	Fills the wall without stretching; the edges that do not fit are cut off.
Original size on canvas	No scaling. The image sits at the top-left corner of the master canvas and each wall shows the part of the image underneath it, like content coming from a media server. Use this when your image was made at canvas resolution.
Original size on wall	No scaling. The image starts at the wall's own top-left corner.

Tile borders & overlays

Set the border colour and thickness, and turn on:

- the tile position arrows (↑ row counted from the bottom, → column from the left);
- tile IDs, such as T05 [C5:R1];
- pixel addresses;
- tile sizes in mm.



Tile borders & overlays

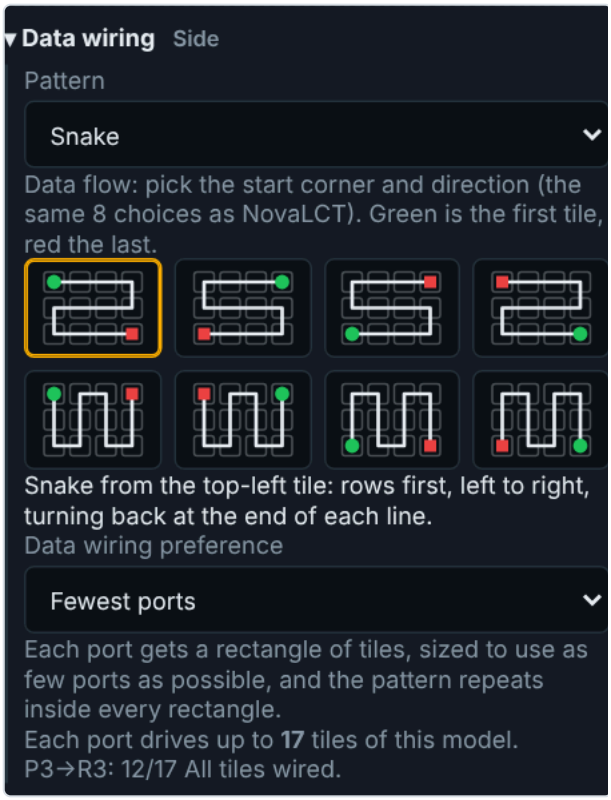


Every overlay on

9. LED processor and data wiring

Choose your **Processor** (Brompton, NovaStar, Megapixel HELIOS, ROE Evision or Custom). The list only offers processors the maker lists for the selected tile (from ROE Visual's brochures and field experience; a tile without published data lists every brand, with a note that its compatibility data is missing). **Receiving card in the tile** narrows it further and **Show incompatible processors** lifts the limit; a processor you already chose stays selected, with a warning naming any wall whose tile does not list it. TileSmith works out the **Pixels per port** for the frame rate and bit depth, and from that how many tiles one port can drive. **Reset to spec** restores the figure after you edit it. Data wiring appears once a processor is chosen.

Data wiring: choose a Pattern (Snake, Linear or Manual) and one of the 8 start icons (green start, red end). Then choose a **Data wiring preference**. **Show data paths** draws the ports on the canvas, P1 to R1 and so on; a port over capacity is drawn dashed red.



Data wiring

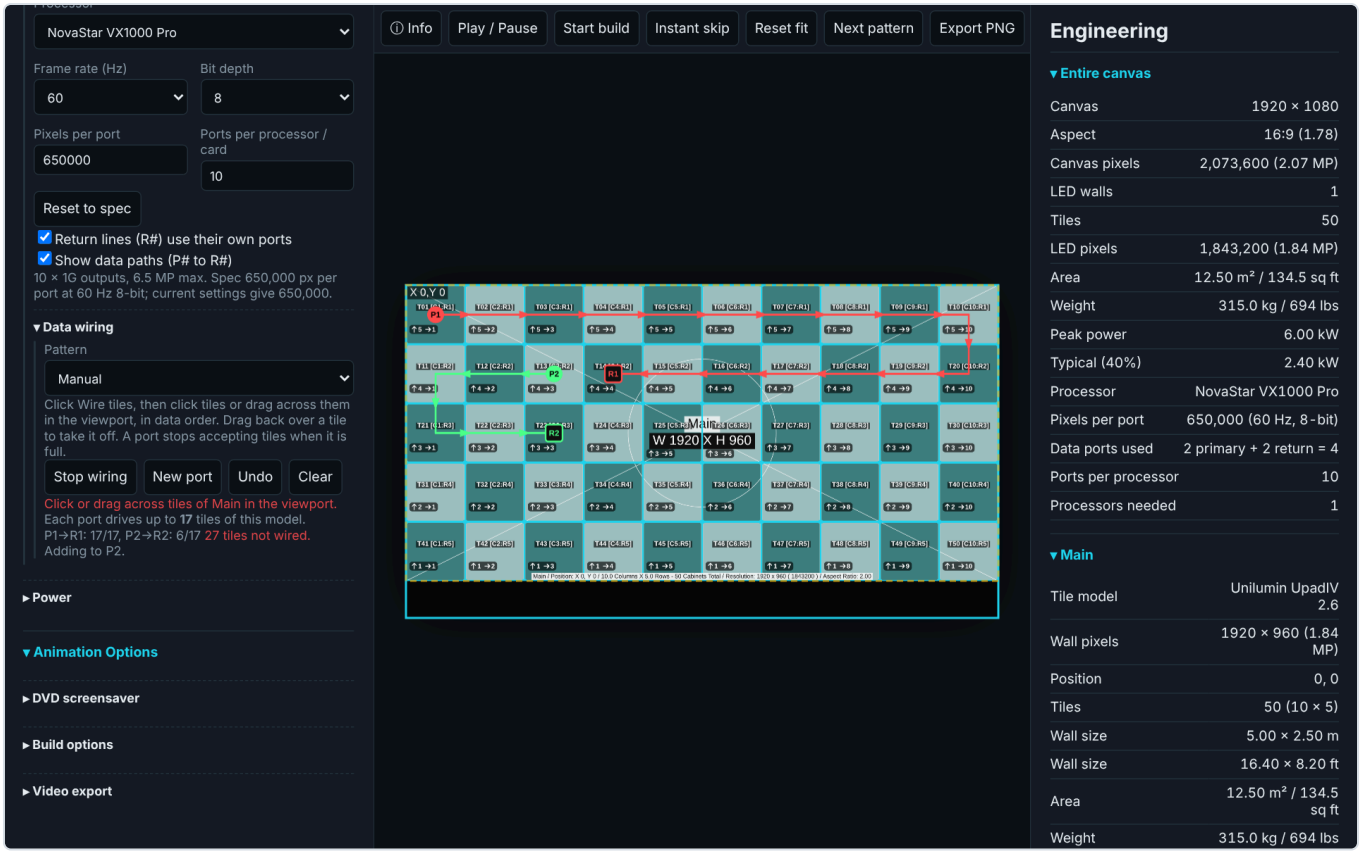
Data wiring preference	How tiles are shared between ports
Fewest ports	Each port gets a rectangle of tiles, sized to use as few ports as possible. The pattern repeats inside every rectangle, like NovaLCT's quick connection.
Whole rows or columns	Each port takes complete rows (complete columns when vertical). If a port cannot drive a whole line, the line is split evenly between ports.
One continuous path	One path runs through the whole wall and is cut wherever a port is full, so a port can start partway along a line.
Horizontal	Each port takes a horizontal band: complete rows across the whole wall, as many as it can drive, whichever way the pattern runs. With a vertical pattern the path snakes up and down inside each band. If a port cannot drive a whole row, each row is split evenly between ports.
Vertical	Each port takes a vertical band: complete columns down the whole wall, as many as it can drive, whichever way the pattern runs. With a horizontal pattern the path snakes back and forth inside each band. If a port cannot drive a whole column, each column is split evenly between ports.



The five preferences on a 10 × 5 wall, 17 tiles per port

Manual wiring

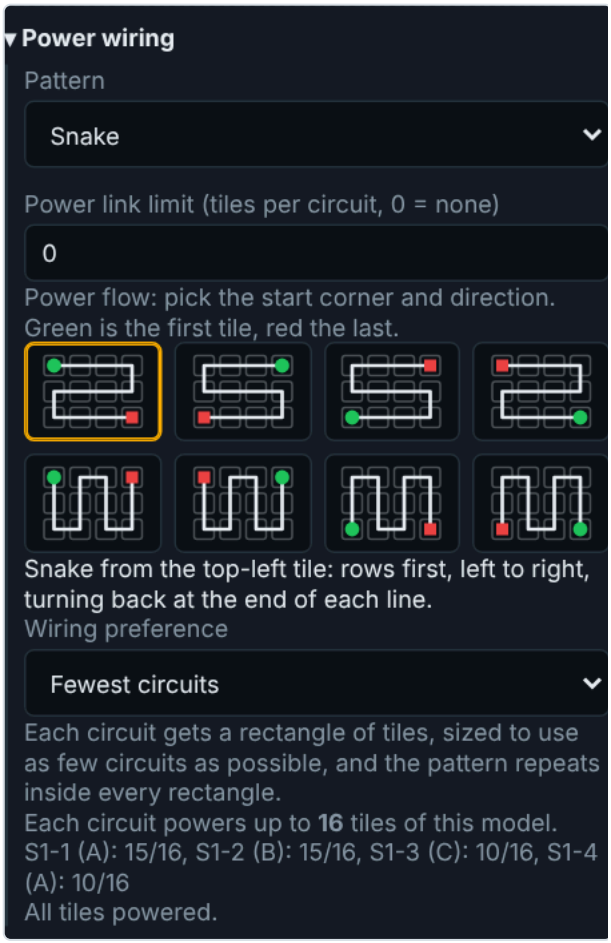
1. Set Pattern to **Manual** and tap **Wire tiles**.
2. Tap tiles in data order, or drag across them. Drag back to take tiles off again; tap the last tile to carry on from it. A port stops taking tiles when it is full.
3. **New port** starts the next port, **Undo** removes the last tile, **Clear** starts again. Tap **Stop wiring** when you are done.



Manual wiring: P1 is full, P2 is being drawn

10. Power and power wiring

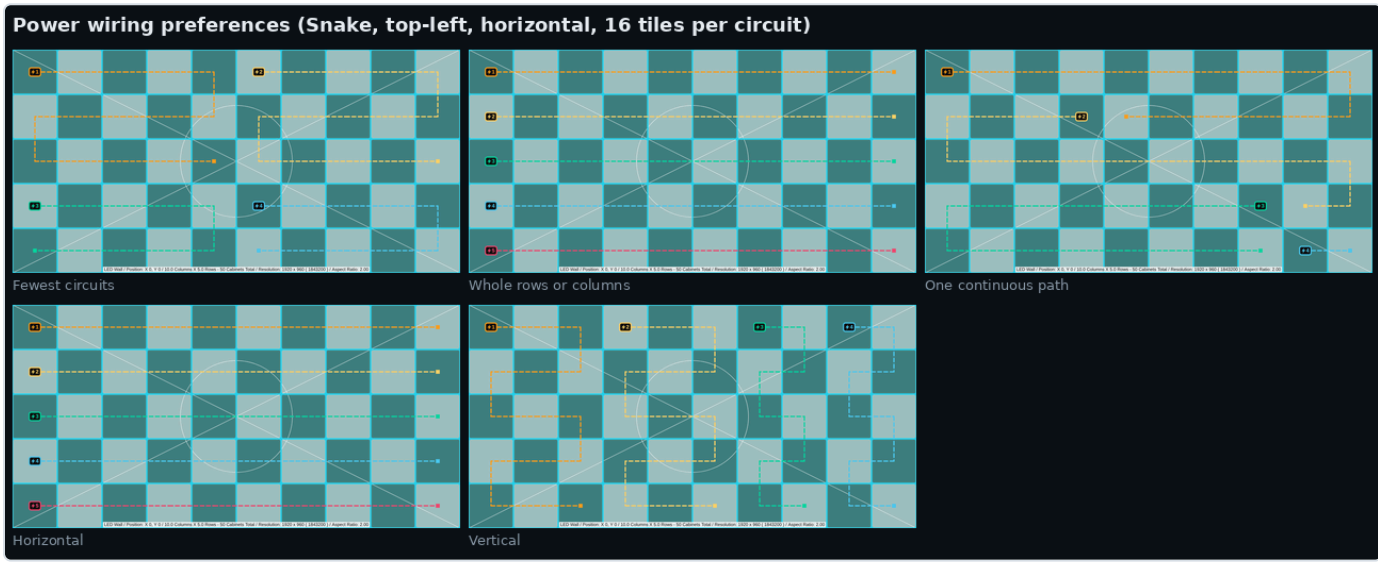
Tick **Power distribution** (Socapex, L14-30, L21-30 or powerCON to Edison cables) to see the power options. Set the circuit voltage, the breaker size and whether circuits are loaded to 80%; TileSmith shows how many tiles one circuit can supply. **Power link limit** caps the tiles per circuit. **Power wiring** works like data wiring, with circuits 1-1 … 1-6, 2-1 … (or 1-X, 1-Y … for twist-lock cables). Phases are left to the distro.



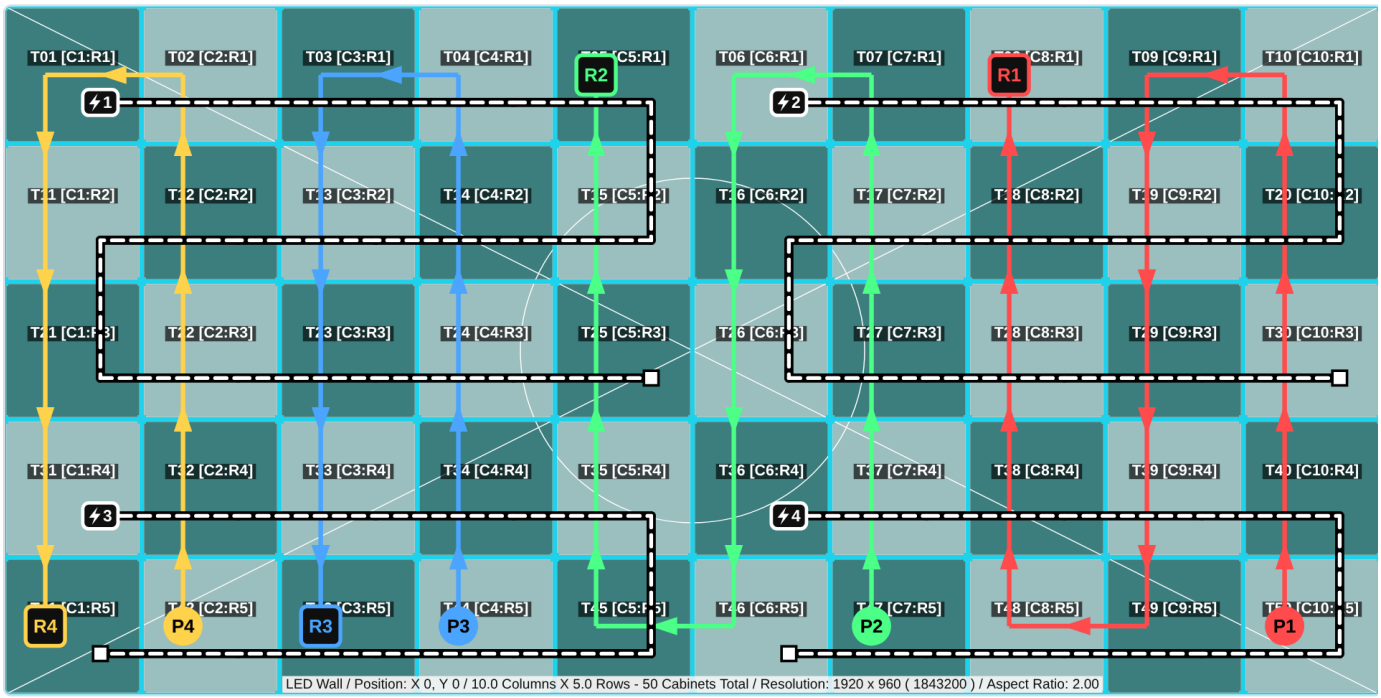
Power wiring

Wiring preference	How tiles are shared between circuits
Fewest circuits	As few power cables as possible, with the tiles shared as evenly as possible between them (circuits may be under-loaded to keep the load balanced) and each circuit a compact block wired as a short snake, so there are no long jumps from side to side. Circuits follow the pattern: rows for Horizontal, columns for Vertical. One extra circuit is only used when it makes every circuit carry the same number of tiles.
Whole rows or columns	Each circuit takes complete rows (complete columns when vertical). If a circuit cannot power a whole line, the line is split evenly between circuits.
One continuous path	One path runs through the whole wall and is cut wherever a circuit is full, so a circuit can start partway along a line.
Horizontal	Each circuit takes a horizontal band: complete rows across the whole wall, as many as it can power, whichever way the pattern runs. If a circuit cannot power a whole row, each row is split evenly between circuits.

Wiring preference	How tiles are shared between circuits
Vertical	Each circuit takes a vertical band: complete columns down the whole wall, as many as it can power, whichever way the pattern runs. If a circuit cannot power a whole column, each column is split evenly between circuits.



The five preferences on a 10 × 5 wall, 16 tiles per circuit

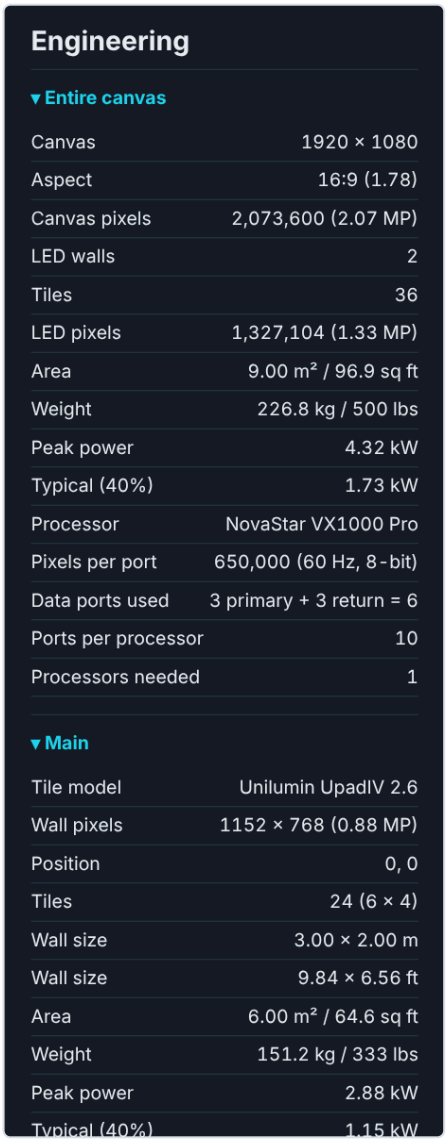


Data and power together: power is drawn white on black, with lightning-bolt badges

11. Engineering data

The **Engineering data** panel (on the right on a tablet; a collapsible menu after the settings on a phone) shows totals for the **Entire canvas** and for each wall. The canvas section covers canvas and LED pixels, area, weight, peak and typical power, processor ports and processors needed, and with power on, circuits, Socapex cables and peak load per phase. Each wall section covers its tile model, wall pixels, position, tiles, size in metres and feet, area, weight, power, tiles per port or circuit, minimum ports and circuits, and its data and power paths.

Tap a section heading to fold it. The same tables go into the spec sheet pages of the exported PDF.



Engineering

12. Build animation and technician

Under **LED build** tap **Start** and a technician builds each wall tile by tile. He walks in carrying the tile, lifts it into its slot and locks it with a flash, then walks off while the next crew member brings the next tile. Each tile takes about 4.8 seconds at Speed 1. Several walls are built at the same time, each starting 0.6 s after the one before.



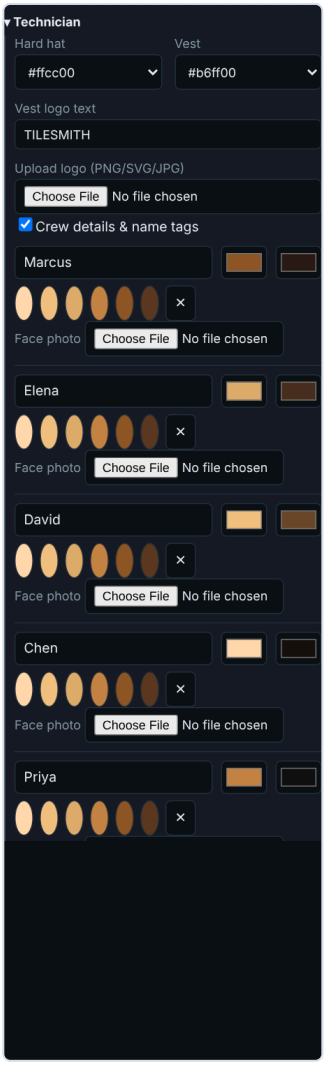
One tile, every 0.2 s

LED build

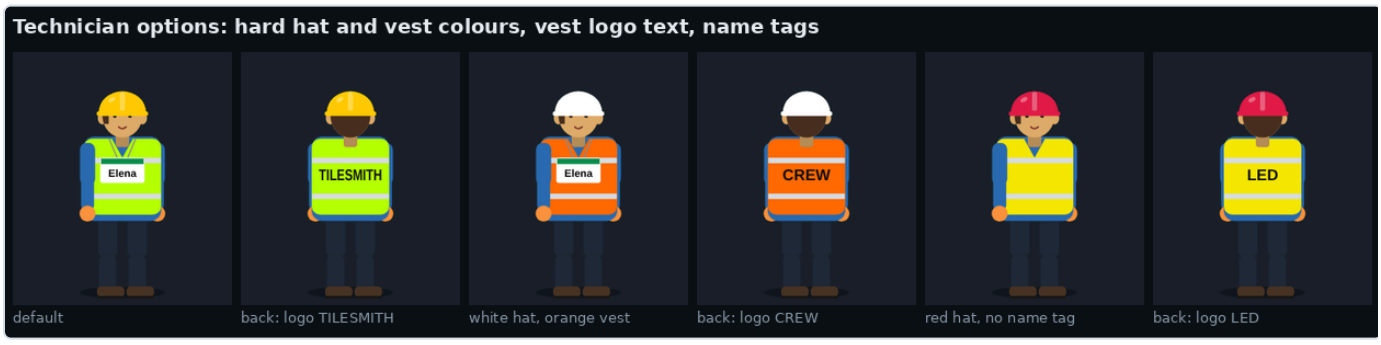
Start, **Stop** (shows the finished walls), **Reset** and **Play / Pause**. **Speed** (0.25× to 5×). **Loop build** restarts after **Pause before restart** seconds, which is handy on a display.

Technician

Pick a **Character style**: Classic or Character pack (the built-in Khalid, Yuki, Kenji and Maria, or your own artwork cut into parts to the Character Pack art brief and loaded as a zip). Choose the hard hat and vest colours and the vest logo text, or upload a logo. With **Crew details & name tags** on, each crew member has a name, skin tone, hair colour and an optional face photo, and wears a name tag. The crew take turns.



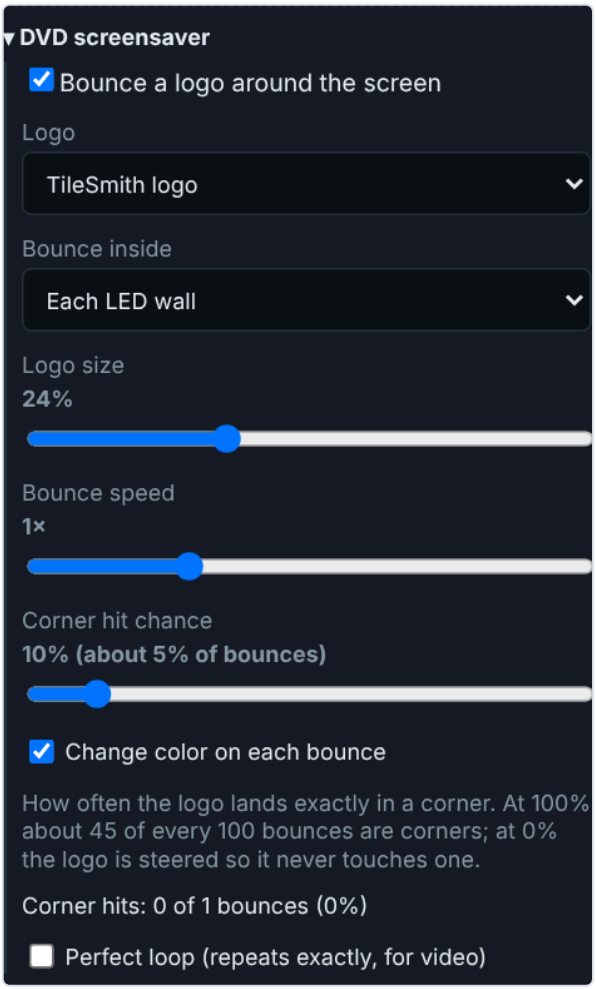
Technician



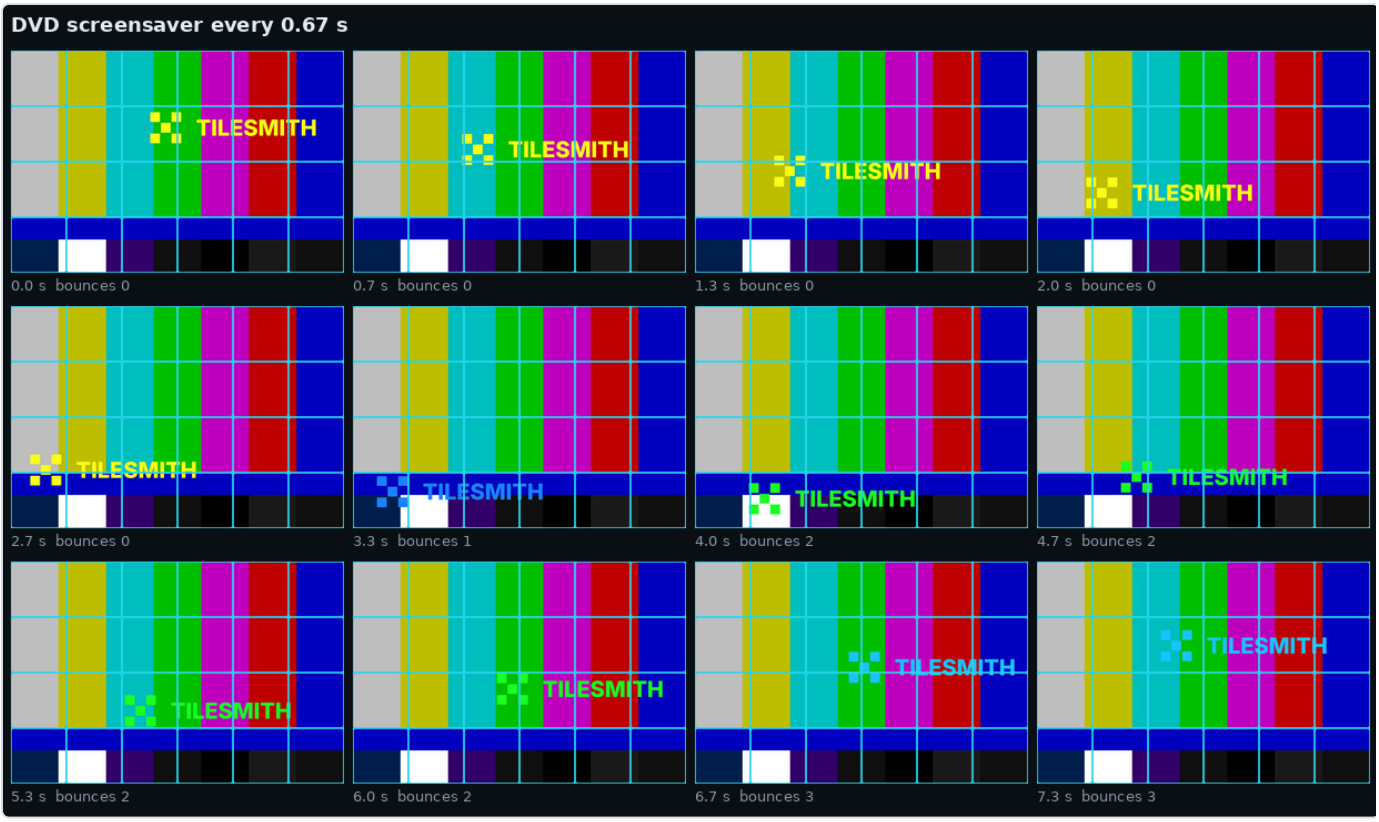
Hard hat, vest and logo options

13. DVD screensaver

Under **Animation Options › DVD screensaver**, a logo bounces inside each wall or across the whole canvas. It can be the TileSmith logo, your text or an uploaded image. Set the size, the speed and the **corner hit chance**. **Change color on each bounce** gives the logo a new colour at every wall. **Perfect loop** makes the motion repeat exactly after the loop length. The loop-video export is a desktop feature.



DVD screensaver

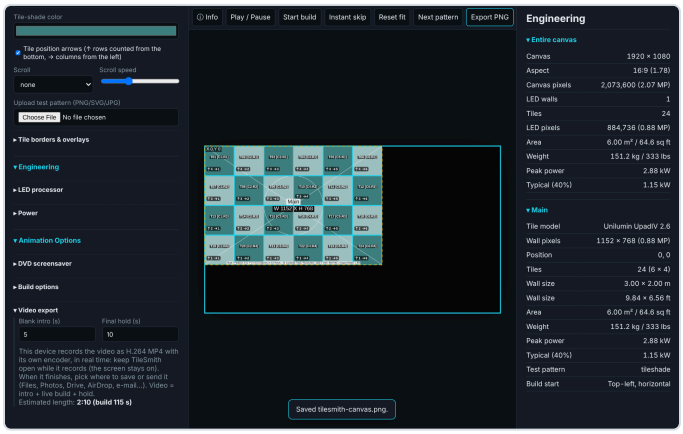


The logo changing colour at each bounce

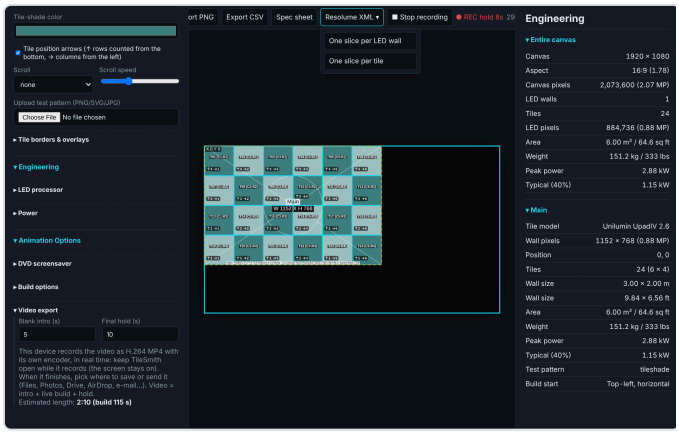
14. Saving and sharing files

Export PNG, Export PDF and Resolume XML open the device's **share sheet**, where you pick where the file goes: Files, Google Drive, iCloud Drive, AirDrop, Quick Share, e-mail, a messaging app and so on.

- When you pick an app, TileSmith says *Saved ….* If you close the sheet, it says *Not saved: …* and nothing is kept.
- **Export PNG**: `tilesmith-canvas.png`, the full canvas at its real size.
- **Export PDF**: `TileSmith-WxH-date.pdf`. Page 1 is the full raster, then the spec sheet with every engineering figure, then the cable labels (one page per label sheet, at your label size). It opens in any PDF viewer.
- **Save label sheet** (Label printing options, shown once data or power wiring is defined): `tilesmith-cable-labels.html`, a sheet of labels for both ends of every data and power cable. Open it in a browser and print it at 100% scale on Avery 5160, 5163, L7160 or roll labels.
- **Resolume XML**: `tilesmith-WxH.xml`. Copy it to *Documents/Resolume Arena/Presets/Advanced Output* on the computer that runs Resolume, then pick it in Arena's Advanced Output.



After Export PNG



Resolume XML menu

15. Recording a video

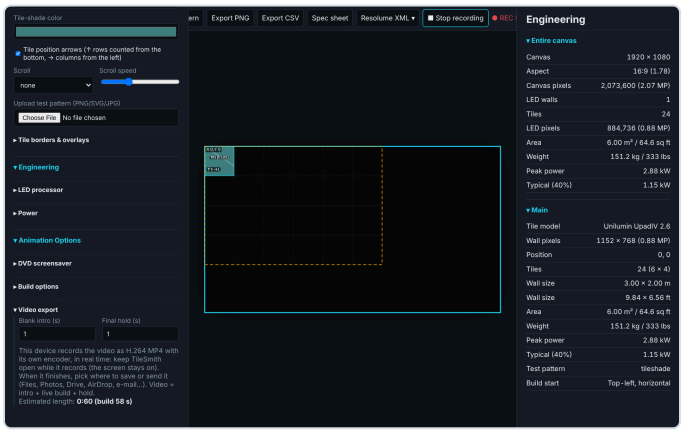
Export video (with **LED build** ticked) records the master canvas as an **H.264 MP4** at 60 frames per second, using the device's own video encoder. The recording has three parts: the **Blank intro**, then the live build, then the **Final hold** of the finished walls. The estimated length is shown under the settings.



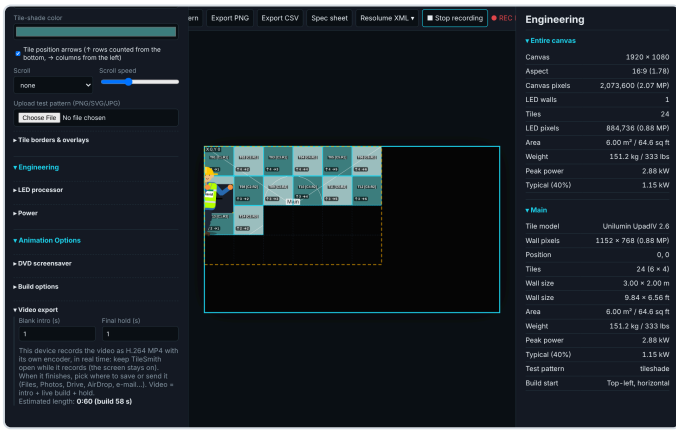
Video export (mobile)

1. Set Blank intro and Final hold (Video export).
2. Tap **Export video**. The top bar shows ● *REC blank*, then *building*, then *hold*.
3. Keep TileSmith open. The screen stays on while it records.
4. When it finishes, the share sheet opens with `tilesmith-build-YYYYMMDD-HHMMSS.mp4`. On iPhone and iPad, *Save Video* puts it in Photos. On Android, choose Photos, Drive or Files.

Tap **Export video** again while it records to cancel.



Recording



Saved

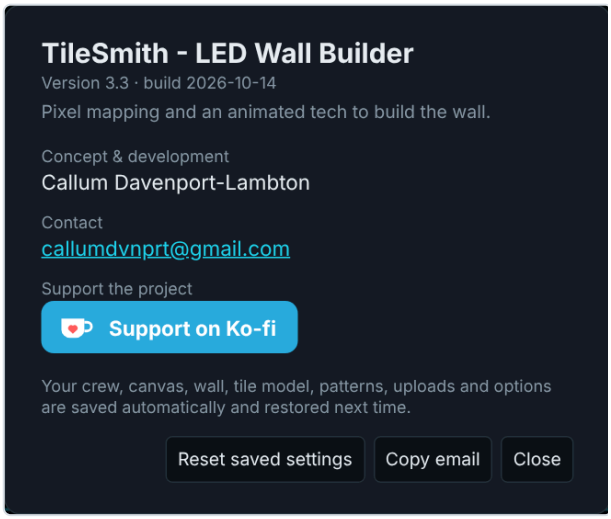
> The video is recorded in real time, so switching apps or locking the device interrupts it. Large canvases on older devices can drop frames. If that happens, tap **Fit canvas to walls** or choose a smaller canvas, and keep the device on its charger. HAP and ProRes need the desktop app.

16. Info, saving and resetting

ⓘ **Info** shows the version, credits, the contact address and the Ko-fi link to support the project. **Reset saved settings** goes back to the defaults after you confirm.

Everything you set up (walls, patterns, wiring, crew, uploaded pictures) is **saved automatically** on the device and comes back the next time you open TileSmith. Installing a newer build keeps it. Uninstalling the app deletes it.

Moving to another device: use **Save** to write a project file and open it on the other device; it carries the walls and wiring plus every custom tile and NovaStar receiving card file the walls use, and adds them to that device's library. A full Backup and Restore feature for all settings is planned for a later version.



Info

17. Battery and speed tips

- When nothing moves, TileSmith stops drawing (the top bar says *still*). The build animation, the DVD screensaver, sweep and scrolling patterns keep it drawing.
- Big canvases (4K and more) and many walls take more work. **Fit canvas to walls** removes empty canvas.
- For long recordings or a looping display, keep the device on its charger.
- Tablets give the most room. On a phone, landscape shows all three columns at once.

18. Troubleshooting

Problem	What to do
Export video says this device cannot record MP4 video	Android: update **Android System WebView** and **Chrome** from Google Play (Android 7 cannot get a new enough version). iPhone and iPad: update iOS / iPadOS.
A save does nothing, or says *Not saved*	Pick an app in the share sheet (Files, Drive ...). Closing the sheet cancels the save.
The video stopped, froze or is shorter than expected	Keep TileSmith open and on screen while it records. Switching apps or locking the device interrupts the recording.
Pinch zooms the whole page instead of the canvas	Pinch inside the viewport. If the whole page still zooms, the build is missing its mobile setup: rebuild with the project's scripts (npm run android / npm run ios, or prepare-assets.js for the AI Studio build).
The app shows "The TileSmith page is not in this app yet"	AI Studio build: copy android-assets/tilesmith into app/src/main/assets/ (see 00-START-HERE.md in the AI Studio kit).
Everything is tiny on a phone	Use the Zoom slider under the viewport, zoom with two fingers, turn the phone to landscape, or use a tablet.
iPhone/iPad app stops opening after a week	Free Apple ID builds expire after 7 days. Run it again from Xcode (BUILD-IOS.txt).
Settings are wrong after testing	ⓘ Info › Reset saved settings.
No Start NDI® button	NDI®, Spout and Syphon are desktop features. Use the Windows or Mac app for live output.

Build problems are covered in BUILD-ANDROID.txt, BUILD-IOS.txt and the AI Studio kit's 00-START-HERE.md.

19. Credits

TileSmith: concept and development by **Callum Davenport-Lambton** · tilesmithapp@gmail.com · support the project at **ko-fi.com/kingof9x**.

The mobile app is built with Capacitor (Android and iOS) or, for the Google AI Studio build, Kotlin with Jetpack Compose and AndroidX WebKit.

NDI® is a registered trademark of Vizrt NDI AB (ndi.video, tools at ndi.video/tools). NDI® output is part of the desktop app and is not included in the mobile app.